

Extension of the Hammer–Rosen Model to a Gray Two-Temperature System

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1 Extension of the Hammer–Rosen Model to a Gray Two-Temperature System

Hammer and Rosen considered the supersonic Marshak-wave problem under the assumption of complete thermodynamic equilibrium, namely

$$T_R = T_m \equiv T.$$

In that case the radiation energy density is written as

$$E = aT^4,$$

and the material energy equation reduces to a single nonlinear diffusion equation for the material temperature. In their notation, the material internal energy and opacity are assumed to obey power laws of the form

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}.$$

This leads to a nonlinear diffusion equation of the form

$$\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^{4+\alpha}}{\partial x^2},$$

where

$$C = \frac{16\sigma g}{3f(4+\alpha)} \rho^{\mu-\lambda-2}.$$

Hammer and Rosen then solve this equation perturbatively using the small parameter

$$\epsilon = \frac{\beta}{4+\alpha}.$$

Their main result for the heat-front position is

$$x_F^2(t) = \frac{2+\epsilon}{1-\epsilon} CH^{-\epsilon}(t) \int_0^t H(t') dt',$$

where

$$H(t) = T_s^{4+\alpha}(t).$$

The absorbed energy per unit area is, to first order in ϵ ,

$$\mathcal{E}(t) = f\rho^{1-\mu} \frac{x_F(t)H^\epsilon(t)}{1-\epsilon}.$$

These are the one-temperature CTE results of Hammer and Rosen. :contentReference[oaicite:0]index=0

2 Gray Two-Temperature Equations

A more general description is obtained by allowing the radiation and material temperatures to differ:

$$T_R \neq T_m.$$

The gray radiation-diffusion equations are

$$\frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) = c\sigma (aT_m^4 - E), \quad (1)$$

$$\rho \frac{\partial e}{\partial t} = c\sigma (E - aT_m^4), \quad (2)$$

where

$$E = aT_R^4, \quad \sigma = \kappa\rho.$$

Using the same power-law assumptions as Hammer and Rosen,

$$e = fT_m^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT_m^\alpha \rho^{-\lambda},$$

we have

$$\kappa = \frac{1}{g} T_m^{-\alpha} \rho^\lambda,$$

and therefore

$$\sigma = \kappa\rho = \frac{1}{g} T_m^{-\alpha} \rho^{1+\lambda}.$$

Thus

$$\frac{c}{3\sigma} = \frac{cg}{3} \rho^{-1-\lambda} T_m^\alpha.$$

Defining

$$D_0 = \frac{cg}{3} \rho^{-1-\lambda}, \quad S_0 = \frac{c}{g} \rho^{1+\lambda}, \quad A_m = f\rho^{1-\mu},$$

the two-temperature gray equations become

$$\frac{\partial E}{\partial t} - \nabla \cdot (D_0 T_m^\alpha \nabla E) = S_0 T_m^{-\alpha} (aT_m^4 - E), \quad (3)$$

$$A_m \frac{\partial T_m^\beta}{\partial t} = S_0 T_m^{-\alpha} (E - aT_m^4). \quad (4)$$

3 Energy Conservation

Adding the radiation and material equations cancels the emission-absorption coupling term. Therefore,

$$\frac{\partial}{\partial t} (E + A_m T_m^\beta) = \nabla \cdot (D_0 T_m^\alpha \nabla E).$$

In one spatial dimension this becomes

$$\frac{\partial}{\partial t} (E + A_m T_m^\beta) = \frac{\partial}{\partial x} \left(D_0 T_m^\alpha \frac{\partial E}{\partial x} \right).$$

The total areal energy inside the heated region is therefore

$$\mathcal{E}(t) = \int_0^{x_F(t)} [E(x, t) + A_m T_m^\beta(x, t)] dx.$$

The incoming flux at the surface is

$$F_{\text{in}}(t) = - D_0 T_m^\alpha \frac{\partial E}{\partial x} \Big|_{x=0}.$$

Hence the exact energy-conservation relation is

$$\boxed{\frac{d\mathcal{E}}{dt} = F_{\text{in}}(t)}$$

or explicitly,

$$\boxed{\frac{d}{dt} \int_0^{x_F(t)} [E(x, t) + A_m T_m^\beta(x, t)] dx = - D_0 T_m^\alpha \frac{\partial E}{\partial x} \Big|_{x=0}.$$

4 Recovery of the Hammer–Rosen Limit

The Hammer–Rosen equation is recovered when the coupling between radiation and matter is sufficiently strong, so that

$$E \simeq a T_m^4.$$

Equivalently,

$$T_R \simeq T_m \equiv T.$$

In this case, the total energy equation becomes

$$\frac{\partial}{\partial t} (a T^4 + A_m T^\beta) = \frac{\partial}{\partial x} \left(D_0 T^\alpha \frac{\partial a T^4}{\partial x} \right).$$

If the material energy dominates the radiation energy,

$$A_m T^\beta \gg a T^4,$$

then

$$A_m \frac{\partial T^\beta}{\partial t} = \frac{\partial}{\partial x} \left(D_0 T^\alpha \frac{\partial a T^4}{\partial x} \right).$$

Since

$$T^\alpha \frac{\partial T^4}{\partial x} = \frac{4}{4+\alpha} \frac{\partial T^{4+\alpha}}{\partial x},$$

we obtain

$$A_m \frac{\partial T^\beta}{\partial t} = \frac{4aD_0}{4+\alpha} \frac{\partial^2 T^{4+\alpha}}{\partial x^2}.$$

Therefore,

$$\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^{4+\alpha}}{\partial x^2},$$

with

$$C = \frac{4aD_0}{(4+\alpha)A_m} = \frac{4acg}{3(4+\alpha)f} \rho^{\mu-\lambda-2}.$$

This has exactly the same mathematical structure as the Hammer–Rosen diffusion equation.

5 Generalized CTE Form Including Radiation Energy

If we still assume

$$T_R = T_m = T,$$

but do not neglect the radiation energy aT^4 , then the conserved energy is

$$U(T) = A_m T^\beta + aT^4.$$

The diffusion variable is

$$\Phi(T) = \int_0^T D_0 T'^\alpha \frac{d(aT'^4)}{dT'} dT'.$$

Since

$$\frac{d(aT'^4)}{dT'} = 4aT'^3,$$

we get

$$\Phi(T) = 4aD_0 \int_0^T T'^{\alpha+3} dT',$$

and therefore

$$\Phi(T) = \frac{4aD_0}{4+\alpha} T^{4+\alpha}.$$

Let

$$\Phi_s(t) = \Phi(T_s(t)).$$

Following the Hammer–Rosen arbitrary-material-property formulation, the heat front may be written as

$$x_F^2(t) \simeq \frac{2}{L_s(t)} \int_0^t \Phi_s(t') dt',$$

where

$$L_s(t) = \frac{2}{\Phi_s(t)} \int_0^{\Phi_s(t)} U(\Phi) \left(1 - \frac{\Phi}{\Phi_s(t)}\right) d\Phi.$$

The corresponding total areal energy is

$$\mathcal{E}(t) \simeq x_F(t) \frac{1}{\Phi_s(t)} \int_0^{\Phi_s(t)} U(\Phi) d\Phi.$$

For the power-law form

$$U(T) = A_m T^\beta + a T^4,$$

define

$$\epsilon_m = \frac{\beta}{4 + \alpha}, \quad \epsilon_r = \frac{4}{4 + \alpha}.$$

Then

$$L_s = 2 \left[\frac{A_m T_s^\beta}{(1 + \epsilon_m)(2 + \epsilon_m)} + \frac{a T_s^4}{(1 + \epsilon_r)(2 + \epsilon_r)} \right].$$

Similarly,

$$\mathcal{E}(t) \simeq x_F(t) \left[\frac{A_m T_s^\beta}{1 + \epsilon_m} + \frac{a T_s^4}{1 + \epsilon_r} \right].$$

6 True Two-Temperature Case

For the full gray system, where

$$T_R \neq T_m,$$

one can still derive an exact energy-conservation equation, but the problem is not closed by energy conservation alone.

Introduce the normalized variables

$$y = \frac{x}{x_F(t)}, \quad E(x, t) = a T_s^4(t) R(y, t), \quad T_m(x, t) = T_s(t) M(y, t).$$

Then the total energy is

$$\mathcal{E}(t) = x_F(t) \left[a T_s^4(t) I_R(t) + A_m T_s^\beta(t) I_M(t) \right],$$

where

$$I_R(t) = \int_0^1 R(y, t) dy, \quad I_M(t) = \int_0^1 M^\beta(y, t) dy.$$

The incoming radiation flux is

$$F_{\text{in}}(t) = \frac{D_0 a T_s^{4+\alpha}(t)}{x_F(t)} A_R(t),$$

where

$$A_R(t) = - M^\alpha(y, t) \frac{\partial R}{\partial y} \Big|_{y=0}.$$

Therefore, the exact moving-front energy equation is

$$\boxed{\frac{d}{dt} \{x_F(t) [a T_s^4(t) I_R(t) + A_m T_s^\beta(t) I_M(t)]\} = \frac{D_0 a T_s^{4+\alpha}(t)}{x_F(t)} A_R(t).}$$

This equation is exact, but it is not closed because the quantities

$$I_R(t), \quad I_M(t), \quad A_R(t)$$

depend on the unknown two-temperature profiles $R(y, t)$ and $M(y, t)$.

Thus, in the true two-temperature case, a closed analytic expression for $x_F(t)$ analogous to the Hammer–Rosen result generally requires solving a new coupled eigenvalue or self-similar problem. Only in the near-equilibrium limit,

$$T_R \simeq T_m,$$

does the system reduce to a Hammer–Rosen-type single-temperature diffusion problem.

7 Henyey-Type Two-Temperature Ansatz

We now ask whether the two-temperature gray system can be reduced by assuming Henyey-like spatial profiles for the radiation and material fields. The gray equations are

$$\frac{\partial E}{\partial t} - \frac{\partial}{\partial x} \left(D_0 T_m^\alpha \frac{\partial E}{\partial x} \right) = S_0 T_m^{-\alpha} (a T_m^4 - E), \quad (5)$$

$$A_m \frac{\partial T_m^\beta}{\partial t} = S_0 T_m^{-\alpha} (E - a T_m^4), \quad (6)$$

where

$$D_0 = \frac{cg}{3} \rho^{-1-\lambda}, \quad S_0 = \frac{c}{g} \rho^{1+\lambda}, \quad A_m = f \rho^{1-\mu}.$$

We introduce the moving-front coordinate

$$y = \frac{x}{x_F(t)}, \quad 0 \leq y \leq 1.$$

We allow the radiation and material temperatures to differ:

$$T_R(x, t) = T_{R,s}(t) \mathcal{R}(y), \quad T_m(x, t) = T_s(t) M(y),$$

and define

$$E(x, t) = E_s(t) R(y), \quad R(y) \equiv \mathcal{R}^4(y),$$

where

$$E_s(t) = aT_{R,s}^4(t).$$

A Henyey-like profile may be written as

$$R(y) = (1 - y)(1 + \chi_R y),$$

and, for the material energy profile,

$$M^\beta(y) = (1 - y)(1 + \chi_M y).$$

The constants χ_R and χ_M describe the curvature of the radiation and material profiles. The Hammer–Rosen first-order Henyey profile has the same structure:

$$U(y) \sim (1 - y)(1 + \chi y).$$

The boundary conditions are

$$R(0) = 1, \quad M(0) = 1, \quad R(1) = 0, \quad M(1) = 0.$$

8 Profile Integrals

Define the radiation and material energy-shape integrals

$$I_R = \int_0^1 R(y) dy, \quad I_M = \int_0^1 M^\beta(y) dy.$$

For the Henyey-like profiles above,

$$I_R = \int_0^1 (1 - y)(1 + \chi_R y) dy = \frac{1}{2} + \frac{\chi_R}{6},$$

and similarly

$$I_M = \frac{1}{2} + \frac{\chi_M}{6}.$$

The surface-gradient coefficient is

$$A_R = - M^\alpha(y) \frac{dR}{dy} \Big|_{y=0}.$$

Since $M(0) = 1$,

$$A_R = -R'(0).$$

For

$$R(y) = (1 - y)(1 + \chi_R y),$$

we have

$$R'(y) = \chi_R - 1 - 2\chi_R y,$$

and therefore

$$R'(0) = \chi_R - 1.$$

Hence

$$\boxed{A_R = 1 - \chi_R.}$$

The incoming radiative flux is therefore

$$F_{\text{in}} = -D_0 T_m^\alpha \frac{\partial E}{\partial x} \Big|_{x=0} = \frac{D_0 T_s^\alpha E_s}{x_F} A_R.$$

9 Total Energy and Heat-Front Equation

The total areal energy inside the heated region is

$$\mathcal{E}(t) = \int_0^{x_F(t)} [E(x, t) + A_m T_m^\beta(x, t)] dx.$$

Using the assumed profiles,

$$\mathcal{E}(t) = x_F(t) [E_s(t) I_R + A_m T_s^\beta(t) I_M].$$

Define

$$B(t) = E_s(t) I_R + A_m T_s^\beta(t) I_M.$$

Then

$$\mathcal{E}(t) = x_F(t) B(t).$$

Energy conservation gives

$$\frac{d}{dt} [x_F(t) B(t)] = \frac{D_0 T_s^\alpha(t) E_s(t)}{x_F(t)} A_R.$$

Define

$$K(t) = D_0 A_R T_s^\alpha(t) E_s(t).$$

Then the moving-front equation becomes

$$\boxed{\frac{d}{dt} [x_F(t) B(t)] = \frac{K(t)}{x_F(t)}}.$$

Multiplying by x_F , we get

$$B x_F \dot{x}_F + x_F^2 \dot{B} = K.$$

Since

$$x_F \dot{x}_F = \frac{1}{2} \frac{dx_F^2}{dt},$$

we obtain

$$\frac{B}{2} \frac{dx_F^2}{dt} + \dot{B} x_F^2 = K.$$

Let

$$X(t) = x_F^2(t).$$

Then

$$\frac{B}{2} \dot{X} + \dot{B} X = K.$$

Dividing by $B/2$,

$$\dot{X} + 2 \frac{\dot{B}}{B} X = \frac{2K}{B}.$$

The integrating factor is

$$\exp \left(\int 2 \frac{\dot{B}}{B} dt \right) = B^2.$$

Therefore,

$$\frac{d}{dt} [B^2(t) X(t)] = 2K(t) B(t).$$

Since $X = x_F^2$, the heat-front position is

$$x_F^2(t) = \frac{2}{B^2(t)} \int_0^t K(t') B(t') dt'.$$

That is,

$$x_F^2(t) = \frac{2}{\left[E_s(t) I_R + A_m T_s^\beta(t) I_M \right]^2} \int_0^t D_0 A_R T_s^\alpha(t') E_s(t') \left[E_s(t') I_R + A_m T_s^\beta(t') I_M \right] dt'.$$

This is the two-temperature Henyey-profile analogue of the Hammer–Rosen front-position formula.

10 Total Energy

Since

$$\mathcal{E}(t) = x_F(t) B(t),$$

we immediately obtain

$$\mathcal{E}^2(t) = 2 \int_0^t K(t') B(t') dt'.$$

Explicitly,

$$\mathcal{E}^2(t) = 2 \int_0^t D_0 A_R T_s^\alpha(t') E_s(t') [E_s(t') I_R + A_m T_s^\beta(t') I_M] dt'.$$

Therefore,

$$\mathcal{E}(t) = \left\{ 2 \int_0^t D_0 A_R T_s^\alpha(t') E_s(t') [E_s(t') I_R + A_m T_s^\beta(t') I_M] dt' \right\}^{1/2}.$$

11 Closure for the Radiation–Material Temperature Difference

The previous result gives a front equation, but the two-temperature problem is not fully closed unless $E_s(t)$ and $T_s(t)$ are known. To close the model, one can integrate the material energy equation over the heated region.

The material energy is

$$\mathcal{E}_m(t) = x_F(t) A_m T_s^\beta(t) I_M.$$

The volume-integrated matter–radiation exchange is

$$Q_{\text{abs}}(t) = \int_0^{x_F} S_0 T_m^{-\alpha} (E - a T_m^4) dx.$$

Using the profile ansatz,

$$Q_{\text{abs}}(t) = x_F S_0 T_s^{-\alpha} [E_s J_R - a T_s^4 J_M],$$

where

$$J_R = \int_0^1 M^{-\alpha}(y) R(y) dy,$$

and

$$J_M = \int_0^1 M^{4-\alpha}(y) dy.$$

Therefore the material-energy equation becomes

$$\frac{d}{dt} [x_F A_m T_s^\beta I_M] = x_F S_0 T_s^{-\alpha} [E_s J_R - a T_s^4 J_M].$$

Similarly, the radiation-energy equation is

$$\frac{d}{dt} [x_F E_s I_R] = \frac{D_0 A_R T_s^\alpha E_s}{x_F} - x_F S_0 T_s^{-\alpha} [E_s J_R - a T_s^4 J_M].$$

Together with

$$x_F^2(t) = \frac{2}{B^2(t)} \int_0^t K(t') B(t') dt',$$

these equations form a closed reduced two-temperature model, provided the shape parameters χ_R and χ_M are prescribed or fitted.

12 Equilibrium Limit

In the complete thermodynamic equilibrium limit,

$$T_R = T_m = T_s,$$

so

$$E_s = aT_s^4.$$

If, in addition, the radiation and material profiles have the same shape, then

$$R(y) \approx M^4(y).$$

The total energy coefficient becomes

$$B(t) = aT_s^4 I_R + A_m T_s^\beta I_M.$$

If material energy dominates radiation energy,

$$A_m T_s^\beta I_M \gg aT_s^4 I_R,$$

then

$$B(t) \approx A_m T_s^\beta I_M.$$

Also,

$$K(t) = D_0 A_R a T_s^{4+\alpha}.$$

Therefore,

$$x_F^2(t) \approx \frac{2D_0 A_R a}{A_m I_M} T_s^{-2\beta}(t) \int_0^t T_s^{4+\alpha+\beta}(t') dt'.$$

This has the same diffusion-wave character as the Hammer–Rosen result, but with constants determined by the chosen Henyey-like profile.